Lec-04

Friday, 4 September 2026 10:58 pm

Iteration for Solving $x = g(x)$

A fundamental principle in computer science is *iteration*. As the name suggests, a process is repeated until an answer is achieved. Iterative techniques are used to find roots of equations, solutions of linear and nonlinear systems of equations, and solutions of differential equations. In this section we study the process of iteration using repeated substitution.

A rule or function $g(x)$ for computing successive terms is needed, together with a starting value p_0. Then a sequence of values $\{p_k\}$ is obtained using the iterative rule $p_{k+1} = g(p_k)$. The sequence has the pattern

$$
\begin{aligned}
&p_0 && \text{(starting value)} \\
&p_1 = g(p_0) \\
&p_2 = g(p_1) \\
&\quad\vdots \\
&p_k = g(p_{k-1}) \\
&p_{k+1} = g(p_k) \\
&\quad\vdots
\end{aligned}
$$

What can we learn from an unending sequence of numbers? If the numbers tend to a limit, we feel that something has been achieved. But what if the numbers diverge or are periodic? The next example addresses this situation.

Example 2.1. The iterative rule $p_0 = 1$ and $p_{k+1} = 1.001 p_k$ for $k = 0, 1, \ldots$ produces a divergent sequence. The first 100 terms look as follows:

$$
\begin{aligned}
p_1 &= 1.001 p_0 &&= (1.001)(1.000000) = 1.001000, \\
p_2 &= 1.001 p_1 &&= (1.001)(1.001000) = 1.002001, \\
p_3 &= 1.001 p_2 &&= (1.001)(1.002001) = 1.003003, \\
&\ \vdots && \quad\vdots \qquad\qquad\qquad \vdots \\
p_{100} &= 1.001 p_{99} &&= (1.001)(1.104012) = 1.105116.
\end{aligned}
$$

The process can be continued indefinitely, and it is easily shown that $\lim_{n\to\infty} p_n = +\infty$. In Chapter 9 we will see that the sequence $\{p_k\}$ is a numerical solution to the differential equation $y' = 0.001y$. The solution is known to be $y(x) = e^{0.001x}$. Indeed, if we compare the 100th term in the sequence with $y(100)$, we see that $p_{100} = 1.105116 \approx 1.105171 = e^{0.1} = y(100)$. ■

In this section we are concerned with the types of functions $g(x)$ that produce convergent sequences $\{p_k\}$.

$$p_{100} \approx 1.105,$$

$$p_{500} \approx 1.647,$$

$$p_{1000} \approx 2.717,$$

$$p_{5000} \approx 148.0.$$

$$p_{k+1} = r p_k, \qquad p_0 \neq 0,$$

$$p_k = p_0 r^k.$$

$$\begin{cases} |r| < 1 & \Rightarrow p_k \to 0 \quad \text{(converges)}, \\ r > 1 & \Rightarrow |p_k| \to \infty \quad \text{(diverges)}, \\ r = 1 & \Rightarrow p_k = p_0 \quad \text{(converges)}, \\ r = -1 & \Rightarrow \text{generally oscillates (diverges)}. \end{cases}$$

Fixed-Point Iteration

A *fixed point* for a function is a number at which the value of the function does not change when the function is applied.

The number p is a **fixed point** for a given function g if $g(p) = p$.

In this section we consider the problem of finding solutions to fixed-point problems and the connection between the fixed-point problems and the root-finding problems we wish to solve. Root-finding problems and fixed-point problems are equivalent classes in the following sense:

- Given a root-finding problem $f(p) = 0$, we can define functions g with a fixed point at p in a number of ways, for example, as

$$g(x) = x - f(x) \quad \text{or as} \quad g(x) = x + 3f(x).$$

- Conversely, if the function g has a fixed point at p, then the function defined by

$$f(x) = x - g(x)$$

has a zero at p.

Finding Fixed Points

Definition 2.1. A ***fixed point*** of a function $g(x)$ is a real number P such that $P = g(P)$. ▲

Geometrically, the fixed points of a function $y = g(x)$ are the points of intersection of $y = g(x)$ and $y = x$.

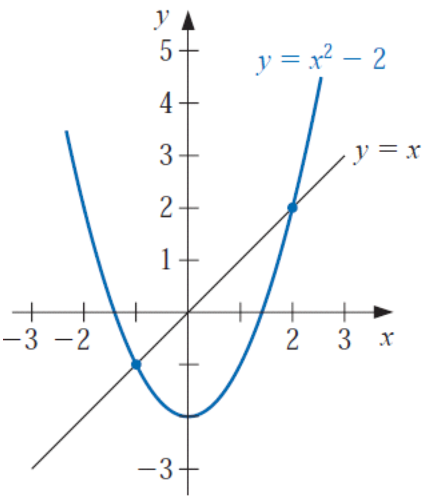
Definition 2.2. The iteration $p_{n+1} = g(p_n)$ for $n = 0, 1, \ldots$ is called ***fixed-point iteration***. ▲

Example 1 Determine any fixed points of the function $g(x) = x^2 - 2$.

Solution A fixed point p for g has the property that

$$p = g(p) = p^2 - 2 \quad \text{which implies that} \quad 0 = p^2 - p - 2 = (p+1)(p-2).$$

A fixed point for g occurs precisely when the graph of $y = g(x)$ intersects the graph of $y = x$, so g has two fixed points, one at $p = -1$ and the other at $p = 2$. These are shown in Figure 2.3. ■



Theorem 2.1. Assume that g is a continuous function and that $\{p_n\}_{n=0}^{\infty}$ is a sequence generated by fixed-point iteration. If $\lim_{n\to\infty} p_n = P$, then P is a fixed point of $g(x)$.

Example 2.2. Consider the convergent iteration

$$p_0 = 0.5 \quad \text{and} \quad p_{k+1} = e^{-p_k} \quad \text{for } k = 0, 1, \ldots.$$

The first 10 terms are obtained by the calculations

$$\begin{aligned}
p_1 &= e^{-0.500000} = 0.606531 \\
p_2 &= e^{-0.606531} = 0.545239 \\
p_3 &= e^{-0.545239} = 0.579703 \\
&\ \ \vdots \qquad\qquad\quad \vdots \\
p_9 &= e^{-0.566409} = 0.567560 \\
p_{10} &= e^{-0.567560} = 0.566907
\end{aligned}$$

The sequence is converging, and further calculations reveal that

$$\lim_{n\to\infty} p_n = 0.567143\ldots.$$

$$x^2 - x - 1 = 0$$

$$x^2 = x + 1$$
$$x = 1 + \frac{1}{x}$$
$$x_{n+1} = 1 + \frac{1}{x_n}$$
Pick $x_1 = 2$
$$x_2 = 1 + \frac{1}{2} = 1.5$$
$$x_3 = 1 + \frac{1}{1.5} = 1.6666$$
$$x_4 = 1 + \frac{1}{1.6666} = 1.6$$
$$x_5 = 1 + \frac{1}{1.6} = 1.625$$
$$x_6 = 1 + \frac{1}{1.625} = 1.612538462$$
Converging to 1.618 ...

$$x^2 - x = 1$$
$$x(x - 1) = 1$$
$$x = \frac{1}{x - 1}$$
$$x_{n+1} = \frac{1}{x_n - 1}$$
Pick $x_1 = 1.6$
$$x_2 = \frac{1}{1.6 - 1} = 1.6666$$
$$x_3 = \frac{1}{1.6666 - 1} = 1.5$$
$$x_4 = \frac{1}{1.5 - 1} = 2$$
$$x_5 = \frac{1}{2 - 1} = 1$$
Not converging

When does it converge?

$$x_{n+1} = \ ...$$
$$x_{n+1} = g(x_n)$$
$$f(root) = 0$$
$$x_{n+1} - root = g(x_n) - g(root)$$
Expand $g(x_n)$ using Taylor Series
$$error_{n+1} = [g(root) + g'(\varphi)(x_n - root)] - g(root)$$
$$error_{\mathrm{n+1}} = g'(\varphi) \times error_n$$
$$|error_{\mathrm{n+1}}| \le |g'(\varphi)||error_n|$$

If $|g'(root)| < 1 \rightarrow$ Converges

Convergence of the examples

$$x_{n+1} = 1 + \frac{1}{x_n}$$

$$g(x) = 1 + \frac{1}{x}$$

$$g'(x) = -\frac{1}{x^2}$$

$$g'\left(\frac{1+\sqrt{5}}{2}\right) = -\frac{1}{\left(\frac{1+\sqrt{5}}{2}\right)^2}$$

$$= -0.3819660112501$$

$$|-0.3819660112501| < 1$$

Converges

$$x_{n+1} = \frac{1}{x_n - 1}$$

$$g(x) = \frac{1}{x - 1}$$

$$g'(x) = -\frac{1}{(x-1)^2}$$

$$g'\left(\frac{1+\sqrt{5}}{2}\right)$$

$$= -\frac{1}{\left(\left(\frac{1+\sqrt{5}}{2}\right) - 1\right)^2}$$

$$= -2.6180339887499$$